
COURSE SYLLABUS
UMBC ENCH 444: Process Engineering Economics and Design I
Fall 2020

Please see [this Google doc](#) for UMBC Policies and Resources during COVID-19

Class: Mon Wed Fri
10.00 – 10.50, 11.00 - 11.50 AM
WEB

Instructor: Gautom Das, Ph.D.
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Office Hours: Tuesdays 10.15 -11.45 AM (please confirm by emails)
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Prerequisites: ENCH 427, ENCH 440
ENCH 445 (can be taken concurrently)

Goals: The learning objectives are as follows:

- To have more in-depth understanding of the basic concepts underlying the design and operation of a chemical process to produce a chemical of interest.
- To perform quantitative calculations related to the design process to meet desired needs within realistic constraints such as economic, environmental, health and safety.
- To work in small teams to demonstrate their fundamental engineering knowledge, research capabilities and resourcefulness while developing a design project.
- To be proficient to use a chemical process simulator Aspen Plus to design a process of interest.
- To consider the professional and ethical responsibilities in engineering design.

Outline of Topics:

Overall the course is about applications of engineering, economic, environmental, and ethical principles in preliminary process design using computer aids such as process simulators. Problem definition literature survey, flowsheet development, material and energy balances, equipment design, profitability analysis, oral and written communication. The course will cover the following topics:

- General Design Considerations
- Process Design Development
- The Structure and Synthesis of Process Flow Diagram
- Flowsheet Synthesis & Development

- Material Transfer and Handling Equipment – Design & Costs
- Reactor, Heat Transfer, and Separation Equipment – Design & Costs
- Materials of Construction
- Capital and Manufacturing Cost Estimation
- Interest, Time Value of Money, Depreciation, Taxes, and Fixed Charges
- Profitability Analysis, Alternative Investments, and Replacements
- Software Used in Process Design: Aspen Plus
- Process optimization: Heat Integration

Textbook and Reading: R Turton, RC Bailie, WB Whiting, JA Shaeiwitz, and D Bhattacharyya, *Analysis, Synthesis and Design of Chemical Processes*, Prentice Hall, 4th Ed., 2012.

Hand-outs and/or supplemental materials will be posted on Blackboard.

There will be reading assignment every week. Reading assignments will be posted on Blackboard. Students are expected to read the assignment to prepare for in class quizzes.

Topics and reading assignments are subject to revision based on progression of the class.

Other References: MS Peters, KD Timmerhaus, and RE West, *Plant Design and Economics for Chemical Engineers*, McGraw-Hill, 5th Ed., 2003.

WD Seider, JD Seader, DR Lewin and S Widagdo, *Process Design Principles : Synthesis, Analysis and Evaluation*, Wiley, 4th Ed, 2017.

(Others recommended books)

Chemical engineering transport texts such as:

J.R. Welty, C.E. Wicks, R.E. Wilson. *Fundamentals of Momentum, Heat, and Mass Transfer*. (any edition) New York, NY: John Wiley & Sons.

R.W. Fox, A.T. McDonald. *Introduction to Fluid Mechanics*. (any edition) New York, NY: John Wiley & Sons.

A chemical engineering kinetics text such as:

H.S. Fogler. *Elements of Chemical Reaction Engineering*. (any edition) Upper Saddle River, NJ: Prentice Hall.

Chemical engineering unit operations text such as:

W.L. McCabe, J.C. Smith, P. Harriott. *Unit Operations of Chemical Engineering*. (any edition) New York, NY: McGraw-Hill.

Course Format & Learning Activities:

Lectures:

- Lectures will complement readings from your text and class handouts. In general, lectures will be interactive, combining in-class discussions with small group problem-solving exercises, chalkboard problem analyses, working on simulation etc.

Quizzes

- Short in-class quizzes (15-20 minutes) will be held in some weeks (see [the Schedule](#) for the dates).

- The quizzes will cover basic concepts found in the topic taught for that particular week and will evaluate your working understanding of the material. Quiz problems will test your understanding of the materials and your contribution to the group project. Therefore, in order to obtain full credit for quizzes, you must keep up with the reading/assignments.
- Students do not take quizzes on the scheduled dates will not receive credit (there are no make-up quizzes). If you cannot take a quiz due to medical or other emergencies, you must notify the instructor in writing (via email).
- The lowest quiz score (one quiz) will be dropped.

Homework

- Homework assignments will be assigned throughout the semester. Please see the [Schedule](#) for dates and topics of the homeworks (*please note dates are subjected to change depends upon the progress of course*).
- Homeworks will require hand calculation, and/or simulations and/or presentations (e.g. on VoiceThread). For longer homeworks students may be given opportunity to team up with one-two other students to complete the work.
- Homeworks must be submitted through Blackboard and/or VoiceThread (as instructed).
- Late submission will not be accepted.

Midterm Examinations:

- There will be one midterm examinations. The exam will be closed book, closed notes; however, students will be allowed to use one page (Letter Size) of notes containing any information that they deem pertinent, with the exception of solved problems.

Final Examination:

- The final exam (2 hours) will be cumulative. Students will be expected to apply understanding of the concepts covered throughout the course to solve comprehensive problems. The final exam will be closed-book; again, students can bring two pages of notes.

Exam Dates:

Midterm Nov 2, in class

Final TBD (as per the final exam schedule given by the Register's Office)

Assessments and Course Grading:

Quizzes	32%
Homeworks	22%
Midterm Exam	16%
Final Exam	30%
TOTAL	100

Course Policies:**Regrading:**

- All requests for regrading must be made via email **within one week of the return** of the assignment in question. The written request should identify clearly the disputed work and justify the request for additional credit. If your work is not clear or the specific question involves subjectivity, then there is no justification for regrading.

Letter Grading:

- **Students with a 90%+ course average will receive an A, an 80-89% average will receive at least a B, 70-79% at least a C, and a 60-69% at least a D.**
- A student cannot pass this course unless the average score in the exams (i.e. average of Midterm and the Final) is 60% or above.
- There will be no make up exams in the class. If you have a professional (e.g., graduate school visit or conference) or UMBC related (e.g., athletic team) reason for missing an exam at its scheduled time, you must provide written documentation for the event. e.g., paid conference registration, graduate school letter. Students unable to attend an exam for valid reasons above can take a second, comprehensive final exam during final exam week (date to be determined). This exam will be counted with the same weight as the exam that was missed. Date & time of second final will be determined later in the semester.
- Dates given for examinations are approximate and might change to account for scheduling considerations.
- Topics and reading assignments are subject to revision based on scheduling considerations.
- In the case of serious emergencies, every effort should be made to notify the instructor prior to the exam. Decisions on allowing the student to take the second final will be based on the validity and verifiability of the excuse. Valid excuses include illness with a doctor's note.

Statement of Values for Student Academic Integrity at UMBC

In February 2001, the Faculty Senate affirmed the importance of our values and practices by adopting the Statement of Values for Student Academic Integrity that is placed on most course syllabi:

Academic integrity is an important value at UMBC. *By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal.*

The purposes of higher education are the learning students and faculty undertake, the knowledge and thinking skills developed, and the enhancement of personal qualities that enable students to be strong contributing members of society. In a competitive world, it is essential that all members of the UMBC community uphold a standard that places integrity of each student's honestly earned achievements above higher grades or easier work dishonestly sought.

All members of the UMBC community are expected to make a commitment to academic honesty in their own actions and with others. Academic misconduct can result in disciplinary action that may include suspension or dismissal. The following are examples of academic misconduct that are not tolerated at UMBC:

- **Cheating:** Using or attempting to use unauthorized material, information, study aids, or another person's work in any academic exercise.
- **Fabrication:** Falsification or invention of any information or citation in an academic exercise.
- **Facilitating academic misconduct:** Helping or attempting to help another student commit an act of academic misconduct.
- **Plagiarism:** Knowingly, or by carelessness or negligence, representing as one's own, in any academic exercise, the intellectual or creative work of someone else.
- **Dishonesty:** Lack of truthfulness or sincerity when interacting with the faculty member regarding an academic exercise

To this end, UMBC undergraduate students also adopted the following Undergraduate Honor Statement as it describes the high standards to which everyone in the community will be held:

I hereby assume the responsibilities of an engaged member in a scholarly and civic community in which academic work and behavior are held to the highest standards of honesty. It is my active participation that affirms these principles and gives them true meaning as well as value in my education. I realize that by committing acts of dishonesty I hurt myself and place an indelible mark on the reputation of UMBC. Therefore, I will not cheat, fabricate materials, plagiarize, or help another to undertake such acts of academic dishonesty, nor will I protect those who engage in acts of academic dishonesty.

For more information on the topic of Academic Integrity, visit: <http://oue.umbc.edu/ai/>

Classes where experiences are shared

Any student who has experienced sexual harassment or assault, relationship violence, and/or stalking is encouraged to seek support and resources. There are a number of resources available to you, which are listed below.

With that said, as an instructor, I am considered a *Responsible Employee*, per UMBC'S interim Policy on Prohibited Sexual Misconduct, Interpersonal Violence, and Other Related Misconduct. This means that while I am here to listen and support you, and I want you to be able to share information related to your life experiences through discussion and written work, I am required to report disclosures of sexual assault, domestic violence, relationship violence, stalking, and/or gender—based harassment to the University's Title IX Coordinator. The purpose of these requirements is for the University to inform you of options, supports, and resources.

You can utilize support and resources even if you do not want to take any further action. You will not be forced to file a police report, but please be aware, depending on the nature of the offense, the University may take action.

If you need to speak with someone in confidence about an incident, UMBC has the following Confidential Resources available to support you:

The Counseling Center: 410-455-2742 (M-F 8:30-5)

University Health Services: 410-455-2542 (M-F 8:30—5)

For after-hours emergency consultation, call the police at 410-455-5555

Other on-campus supports and resources:

The Women's Center (available to students of all genders): 410-455-2714 (M-Th 9:30-6, F 9:30-4)

Title IX Coordinator: 410-455-1606 (9-5)

Child Abuse and Neglect

Please note that Maryland law requires that I report all disclosures or suspicions of child abuse or neglect to the Department of Social Service and/or the police.